Notes on Multi-Agent Traversal and Communication Case Studies

1 Overview

We consider a communication complexity model in which players each own subsets of vertices (and possibly edges) of a known global graph. The task is to compute shortest-path distances between nodes using only vertices belonging to the union of players' subsets. Here are ten case studies where this problem models a real-world situation wherein communication is in some form a bottleneck.

1.1 Volcanic ash shutdown and airline re-routing (Europe, 2010)

Case study. The April–May 2010 Eyjafjallajökull eruption triggered large-scale European airspace closures, disrupting on the order of 10^5 flights and 10^7 passenger journeys. This forced airlines and air-navigation authorities to continuously re-plan feasible routes as airspace sectors opened and closed. [1]

Model. Let each air-navigation service provider or airspace-sector authority be a player owning vertices/edges corresponding to waypoints and sectors. Feasible shortest paths must lie entirely within currently open subsets.

1.2 Airport-wide power failure as localized vertex removal (Atlanta ATL, 2017)

Case study. In December 2017, Hartsfield–Jackson Atlanta International Airport suffered a major power outage that caused widespread cancellations and passenger stranding. Airlines re-routed passengers through alternative hubs as systems were gradually restored. [2]

Model. Airport operators own terminal, gate, and people-mover vertices, while airlines own flight-leg vertices. Shortest-path computation becomes layered (physical movement plus flight network) with partial vertex restoration over time.

1.3 Urban transit shutdown and phased restoration (NYC Hurricane Sandy, 2012)

Case study. Hurricane Sandy caused major disruptions to New York City's subway, commuter rail, and road tunnels, followed by staged service restoration over weeks and months. [3, 4]

Model. Different transit agencies act as players owning disjoint subsets of stations, tunnels, and tracks. Shortest paths are constrained to open infrastructure and evolve as vertices are gradually reintroduced.

1.4 Highway bridge collapse and rapid detour routing (I-95 Philadelphia, 2023)

Case study. A tanker fire on June 11, 2023 caused the collapse of an I-95 overpass in Philadelphia, forcing large-scale detours until phased reopening. [5, 6]

Model. State DOTs, city streets, and toll operators are distinct players owning roadway vertices and edges. The event motivates shortest-path computation after the removal of a critical cut vertex or edge. Perhaps we can make the graph weighted but only players know the weights, simulating traffic.

1.5 Global shipping chokepoint blockage (Suez Canal, 2021)

Case study. The grounding of the Ever Given in March 2021 blocked the Suez Canal for six days, forcing carriers to either wait or reroute around the Cape of Good Hope. [7, 8]

Model. Shipping lines and canal authorities own port calls and chokepoint edges. Shortest paths become time-dependent and depend on proprietary schedule and capacity information.

1.6 Port access loss and cargo diversion (Baltimore Key Bridge collapse, 2024)

Case study. The collapse of the Francis Scott Key Bridge blocked access to the Port of Baltimore, requiring cargo diversion to alternative ports until the channel was reopened in phases. [9, 10]

Model. Port authorities, terminal operators, and carriers own different layers of a sea–port–land graph. This motivates multi-layer shortest-path problems under partial information and staged recovery.

1.7 Electric power grid load shedding (Texas ERCOT, 2021)

Case study. Extreme cold in February 2021 caused generation failures and load shedding across Texas, with cascading risks during restoration. [11, 12]

Model. Utilities and grid operators own substations, lines, and generators. The analogue of shortest paths is determining feasible low-cost delivery paths from supply to demand under connectivity constraints.

1.8 Fuel pipeline cyber disruption (Colonial Pipeline, 2021)

Case study. A ransomware attack in May 2021 shut down the Colonial Pipeline, disrupting fuel logistics across the U.S. East Coast. [13, 14]

Model. Pipeline operators and distributors own depots and transfer vertices. The case highlights adversarial vertex failures and asymmetric information about availability.

1.9 Backbone withdrawal and global reachability failure (Meta, 2021)

Case study. A configuration error in October 2021 withdrew Meta's backbone routes, making major services unreachable until recovery. [15, 16]

Model. Autonomous systems are players owning routing vertices and edges. Shortest-path feasibility corresponds to reachability under route withdrawals with limited policy disclosure.

1.10 Managed DNS attack and service unavailability (Dyn, 2016)

Case study. A large-scale DDoS attack on Dyn in October 2016 disrupted access to many major websites due to DNS resolution failures. [17]

Model. Infrastructure providers own critical dependency vertices. Shortest-path analogues arise in service-composition graphs constrained to available components.

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